

Homework | Multiple-choice and written practice

12 original questions, 30 teacher marks, about 45 minutes. Q5 requires exactly TWO choices. Show predictions, traces and reasons. Braces mark blocks; each statement is on its own line.

AP CSP | Lesson 15: Randomness & Simulation

Written responses | Questions 7-9

7. [4 marks]

For RANDOM(3, 7): (a) list the outcomes; (b) count them; (c) give $P(\text{result} = 7)$; (d) give $P(\text{even result})$, with a reason.

8. [4 marks]

A fictional model needs a 30% chance of Delay each trial. Use $r \leftarrow \text{RANDOM}(1, 10)$.

(a) Give a Delay condition. (b) Justify its probability.

(c) State the other outcome. (d) Explain why $r < 3$ would not work.

9. [4 marks]

```
heads ← 0
REPEAT 4 TIMES
{
  r ← RANDOM(1, 2)
  IF(r = 1)
  {
    heads ← heads + 1
  }
}
```

For supplied results 1, 2, 1, 1: give (a) heads after each trial, (b) final heads, (c) observed fraction, (d) whether the next run must match.

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Written responses | Questions 10-12

10. [4 marks]

A fair independent coin model yields 53 Heads in 100 trials.

- (a) State theoretical $P(\text{Heads})$. (b) Calculate observed frequency.
(c) Must 1,000 trials give exactly 500 Heads? Explain.
(d) Must the next toss correct the earlier imbalance? Explain.

11. [4 marks]

Compare $\text{sum} \leftarrow \text{RANDOM}(1, 6) + \text{RANDOM}(1, 6)$ with $r \leftarrow \text{RANDOM}(1, 6)$, then $\text{sum} \leftarrow 2 * r$.

- (a) State each model's number of draws. (b) Give an odd sum possible only in the first.
(c) Explain the second model's restriction. (d) Which represents two independent fair dice?

12. [4 marks]

A fictional school queue model assigns a 30% delay chance to each independent arrival, regardless of time.

- (a) Give one useful purpose. (b) State one simplifying assumption.
(c) Name an omitted factor and possible effect. (d) Explain why more trials alone cannot repair that omission.
